

# $K$ -theoretic Catalan functions

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- Schubert calculus
- Catalan functions: a new approach to old problems
- $K$ -theoretic Catalan functions

# Overview of Schubert Calculus Combinatorics

## Geometric problem

Find  $c_{\lambda\mu}^\nu = \#$  of points in intersection of subvarieties in a variety  $X$ .

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## Representatives

Special basis of polynomials  $\{f_\lambda\}$  such that  $f_\lambda \cdot f_\mu = \sum_\nu c_{\lambda\mu}^\nu f_\nu$

# Overview of Schubert Calculus Combinatorics (cont.)

Combinatorial study of  $\{f_\lambda\}$  enlightens the geometry (and cohomology).

## Goal

Identify  $\{f_\lambda\}$  in explicit (simple) terms amenable to calculation and proofs.

## Geometric problem

Find  $c_{\lambda\mu}^\nu = \#$  of points in intersection of Schubert varieties  $\{X_\lambda\}_{\lambda \subseteq (n^m)}$  in variety  $X = \text{Gr}(m, n)$ .

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## Representatives

Special basis of Schur polynomials  $\{s_\lambda\}$  such that  $s_\lambda \cdot s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$  for Littlewood-Richardson coefficients  $c_{\lambda\mu}^\nu$ .

## Example

*Semistandard tableaux*: columns increasing and rows non-decreasing.

|   |   |   |   |
|---|---|---|---|
| 5 |   |   |   |
| 3 | 4 |   |   |
| 2 | 3 |   |   |
| 1 | 2 | 2 | 5 |

|   |   |   |   |
|---|---|---|---|
| 8 |   |   |   |
| 7 | 9 |   |   |
| 3 | 4 |   |   |
| 1 | 2 | 5 | 6 |

standard = no repeated letters

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Schur function  $s_\lambda$  is a “weight generating function” of semistandard tableaux:

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 2 | 3 | 3 | 2 | 3 | 3 | 2 | 3 |
| 1 | 1 | 1 | 1 | 1 | 2 | 1 | 2 |

$$s_{\square}(x_1, x_2, x_3) = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_2^2 + x_1 x_3^2 + x_2 x_3^2 + 2x_1 x_2 x_3$$

# Schur functions $s_\lambda$ (cont.)

## Pieri rule

Determines multiplicative structure:

$$s_r s_\lambda = \sum (1 \text{ or } 0) s_\nu$$

$$s_{\square} s_{\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}} = s_{\begin{smallmatrix} \square & \square & \square \\ \square & \end{smallmatrix}} + s_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} + s_{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}}$$

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Since  $s_{\mu_1} \cdots s_{\mu_r} = s_{(\mu_1, \dots, \mu_r)} + \text{lower order terms}$ , subtract to get

$$s_{(\mu_1, \dots, \mu_r)} s_\lambda = \sum c_{\lambda\mu}^\nu s_\nu$$

for well-understood *Littlewood-Richardson coefficients*  $c_{\lambda\mu}^\nu$ .

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$$\mathfrak{S}_{s_i} = x_1 + \cdots + x_i$$

### Open Problem

Structure constants  $\mathfrak{S}_w \mathfrak{S}_u = \sum_v c_{wu}^v \mathfrak{S}_v$  have no tableaux description.

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| Theory                                 | $f_\lambda$                  |
|----------------------------------------|------------------------------|
| (Co)homology of Grassmannian           | Schur functions              |
| (Co)homology of flag variety           | Schubert polynomials         |
| Quantum cohomology of flag variety     | Quantum Schuberts            |
| (Co)homology of Types BCD Grassmannian | Schur- $P$ and $Q$ functions |
| (Co)homology of affine Grassmannian    | (dual) $k$ -Schur functions  |
| $K$ -theory of Grassmannian            | Grothendieck polynomials     |
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And many more!

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$$\begin{aligned}\Phi: QH^*(Fl_{k+1}) &\rightarrow H_*(Gr_{SL_{k+1}})_{loc} \\ \mathfrak{S}_w^Q &\mapsto \frac{s_\lambda^{(k)}}{\prod_{i \in Des(w)} \tau_i}\end{aligned}$$

where  $s_\lambda^{(k)}$  is a  $k$ -Schur symmetric function and  $Gr_{SL_{k+1}}$  is the “affine Grassmannian.”

Upshot

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## Upshot

Computations for (quantum) Schubert polynomials can be moved into symmetric functions.

# $k$ -Schur functions

- $s_{\lambda}^{(k)}$  for  $\lambda_1 \leq k$  a basis for  $\mathbb{Z}[s_1, s_2, \dots, s_k]$  (Lapointe et al., 2003).

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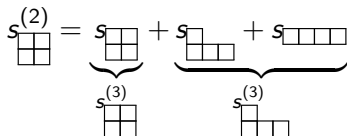
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- Branching with positive coefficients (Lam et al., 2010):

$$s_{\lambda}^{(2)} = s_{\lambda} + s_{\lambda} + s_{\lambda}$$

The diagram shows the branching rule for  $k$ -Schur functions. It illustrates that  $s_{\lambda}^{(2)}$  (a 2x2 square) can be expressed as the sum of three terms. The first term is  $s_{\lambda}^{(3)}$  (a 3x2 rectangle). The second and third terms are  $s_{\lambda}^{(3)}$  (a 3x1 vertical rectangle). Brackets indicate that the second and third terms are grouped together as  $s_{\lambda}^{(3)}$ .

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The diagram shows the branching of the 2-Schur function  $s_{\lambda}^{(2)}$  into three 3-Schur functions. On the left,  $s_{\lambda}^{(2)}$  is represented by a 2x2 square. On the right, it is equal to the sum of three 3-Schur functions:  $s_{\lambda}^{(3)}$  (a 2x2 square),  $s_{\lambda}^{(3)}$  (a 2x1 rectangle), and  $s_{\lambda}^{(3)}$  (a 1x3 horizontal row). Brackets below the right side group the three terms under  $s_{(2)}^{(3)}$ ,  $s_{(1,1)}^{(3)}$ , and  $s_{(1,1,1)}^{(3)}$  respectively.

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- (Lam et al., 2010) gives geometric interpretation,
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- Branching with  $t$  important for Macdonald polynomial positivity.
- Many conjecturally equivalent definitions.

- Schubert calculus
- **Catalan functions: a new approach to old problems**
- $K$ -theoretic Catalan functions

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Key: Catalan functions = large class of symmetric functions.

# Ingredients for Catalan functions

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# Raising Operators on Symmetric Functions

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$$s_{22} = (1 - R_{12})h_{22} = h_{22} - h_{31}$$

$$s_{211} = (1 - R_{12})(1 - R_{23})(1 - R_{13})h_{211}$$

$$= h_{211} - h_{301} - h_{220} - \text{red } h_{310} + \text{red } h_{310} + \underbrace{h_{32-1}}_{=0} + h_{400} - \underbrace{h_{41-1}}_{=0}$$

some terms cancel

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Simplifies formulas. E.g., for  $\langle s_{1^r}^\perp s_\lambda, s_\mu \rangle = \langle s_\lambda, s_{1^r} s_\mu \rangle$ ,

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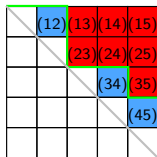
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$$s_{1^r}^\perp s_\lambda = \sum_{S \subseteq [1, \ell], |S|=r} s_{\lambda - \epsilon_S}$$

$$s_{1^2}^\perp s_{333} = s_{322} + s_{232} + s_{223}$$

# Root Ideals

A root ideal  $\Psi$  of type  $A_{\ell-1}$  positive roots: given by Dyck path (lattice path above diagonal).

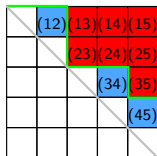


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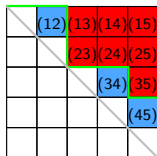
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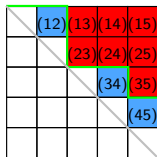
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## Intuition

Catalan functions interpolate between  $h_\lambda$  and  $s_\lambda$ .

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## Theorem (Blasiak et al., 2020)

For  $\Psi$  any root ideal and  $\lambda$  a partition,  $H(\Psi; \lambda)$  is Schur positive!

## $k$ -Schur root ideal for $\lambda$

$$\begin{aligned}\psi = \Delta^k(\lambda) &= \{(i, j) : j > k - \lambda_i\} \\ &= \text{root ideal with } k - \lambda_i \text{ non-roots in row } i\end{aligned}$$

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|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 3 |   |   |   |   |   |
|   | 3 |   |   |   |   |
|   |   | 2 |   |   |   |
|   |   |   | 2 |   |   |
|   |   |   |   | 1 |   |
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$\leftarrow$  row  $i$  has  $4 - \lambda_i$  non-roots

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$k$ -Schur is a Catalan function (Blasiak et al., 2019).

For partition  $\lambda$  with  $\lambda_1 \leq k$ ,

$$s_{\lambda}^{(k)} = H(\Delta^k(\lambda); \lambda).$$

# Key ingredient of branching proof

Dual vertical Pieri rule:  $s_{1^r}^\perp s_\lambda^{(k)} = \sum_\mu a_{\lambda\mu} s_\mu^{(k)}$  for  $\langle s_{1^r}^\perp f, g \rangle = \langle f, s_{1^r} g \rangle$ .

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|   | 3 |   |   |   |   |
|   |   | 2 |   |   |   |
|   |   |   | 2 |   |   |
|   |   |   |   | 1 |   |
|   |   |   |   |   | 1 |

$$\Delta^5(4, 4, 3, 3, 2, 2) =$$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 4 |   |   |   |   |   |
|   | 4 |   |   |   |   |
|   |   | 3 |   |   |   |
|   |   |   | 3 |   |   |
|   |   |   |   | 2 |   |
|   |   |   |   |   | 2 |

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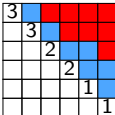
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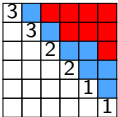
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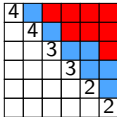
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Branching is a special case of Pieri:

$$s_\lambda^{(k)} = s_{1^\ell}^\perp s_{\lambda+1^\ell}^{(k+1)} = \sum_\mu a_{\lambda+1^\ell, \mu} s_\mu^{(k+1)}$$

- Schubert calculus
- Catalan functions: a new approach to old problems
- **$K$ -theoretic Catalan functions**

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- Inhomogeneous basis:  $g_\lambda = s_\lambda + \text{lower degree terms}$ .

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Add (addable) or mark (removable) in any combination of  $r$  boxes, but only once per row.

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- Dual to Grothendieck polynomials  $G_\lambda$ : Schubert representatives for  $K^*(Gr(m, n))$

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The diagram illustrates the Pieri rule for  $K$ -Schur functions. It shows the product of a 1-strip (represented by a 5x5 grid of colored dots) and a 2-bounded partition (represented by a 5x5 grid of colored dots). The result is the difference of two 2-bounded partitions (represented by 5x5 grids of colored dots). The partitions are labeled as  $g_1 g_{211}^{(2)} = g_{2111}^{(2)} - 2g_{211}^{(2)}$ . The partitions are visualized as 5x5 grids of colored dots (red, blue, black) with some cells shaded gray to indicate the addition or subtraction of cells. The text "2-bounded partitions  $\leftrightarrow$  3-cores" is also present.

- Conjecture:  $g_{\lambda}^{(k)}$  have positive branching into  $g_{\mu}^{(k+1)}$  (Lam et al., 2010; Morse, 2011).

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No direct formula for  $g_{\lambda}^{(k)}$

## Solution

Find a formula for  $g_{\lambda}^{(k)}$  analogous to raising operator formula for  $s_{\lambda}^{(k)}$ .

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Requires an inhomogeneous refinement of Catalan functions.

# An Extra Ingredient: Lowering Operators

Lowering Operators  $L_j(f_\lambda) = f_{\lambda - \epsilon_j}$

$$L_3 \left( \begin{array}{|c|c|c|} \hline \text{red} & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \right) = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array}, \quad L_1 \left( \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \text{red} \\ \hline \end{array} \right) = \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array}$$

## $K$ -theoretic Catalan function

Let  $\Psi, \mathcal{L} \subseteq \Delta_\ell^+$  be order ideals of positive roots and  $\gamma \in \mathbb{Z}^\ell$ , then

$$K(\Psi; \mathcal{L}; \gamma) := \prod_{(i,j) \in \mathcal{L}} (1 - L_j) \prod_{(i,j) \in \Delta_\ell^+ \setminus \Psi} (1 - R_{ij}) k_\gamma$$

# Affine $K$ -Theory Representatives with Raising Operators

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## Example

non-roots of  $\Psi$ , roots of  $\mathcal{L}$

|  |      |      |      |      |
|--|------|------|------|------|
|  | (12) | (13) | (14) | (15) |
|  |      | (23) | (24) | (25) |
|  |      |      | (34) | (35) |
|  |      |      |      | (45) |
|  |      |      |      |      |

$$\begin{aligned} K(\Psi; \mathcal{L}; 54332) \\ = (1 - L_4)^2 (1 - L_5)^2 (1 - R_{12}) (1 - R_{34}) (1 - R_{45}) k_{54332} \end{aligned}$$

Answer (Blasiak-Morse-S., 2020)

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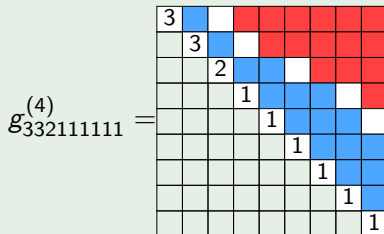
For  $K$ -homology of affine Grassmannian,  $g_{\lambda}^{(k)} = K(\Delta^k(\lambda); \Delta^{k+1}(\lambda); \lambda)$  since this family satisfies the Pieri rule.

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Example



$$\Delta_9^+ / \Delta^4(332111111), \Delta^5(332111111)$$

# Pieri Rule Illustrated (Recurrences)

A “graphical calculus.”

$$g_1 g_{211}^{(2)}$$

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$$g_1 g_{211}^{(2)}$$

=

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
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|   |   |   | 0 |   |   |   |
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# Pieri Rule Illustrated (Straightening)

$$g_1 g_{211}^{(2)} =$$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
|   |   | 1 |   |   |   |   |
|   |   |   | 0 |   |   |   |
|   |   |   |   | 0 |   |   |
|   |   |   |   |   | 0 |   |
|   |   |   |   |   |   | 1 |

 $+$ 

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| 2 |   |   |   |   |   |   |
|   | 1 |   |   |   |   |   |
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|   |   |   | 0 |   |   |   |
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|   |   | 1 |   |   |   |   |
|   |   |   | 1 |   |   |   |
|   |   |   |   | 0 |   |   |
|   |   |   |   |   | 0 |   |
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$$=$$

|   |   |   |   |
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| 2 |   |   |   |
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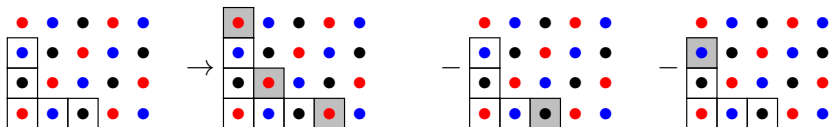
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## Thank you!

- Anderson, David, Linda Chen, and Hsian-Hua Tseng. 2017. *On the quantum  $K$ -ring of the flag manifold*, preprint. arXiv: 1711.08414.
- Blasiak, Jonah, Jennifer Morse, Anna Pun, and Daniel Summers. 2019. *Catalan Functions and  $k$ -Schur Positivity*, J. Amer. Math. Soc. **32**, no. 4, 921–963.
- Blasiak, Jonah, Jennifer Morse, and Anna Pun. 2020. *Demazure crystals and the Schur positivity of Catalan functions*, preprint. arXiv: 2007.04952.
- Blasiak, Jonah, Jennifer Morse, and George H. Seelinger. 2020.  *$K$ -theoretic Catalan functions*, preprint. arXiv: 2010.01759.
- Chen, Li-Chung. 2010. *Skew-linked partitions and a representation theoretic model for  $k$ -Schur functions*, Ph.D. thesis.
- Fomin, Sergey, Sergei Gelfand, and Alexander Postnikov. 1997. *Quantum Schubert polynomials*, J. Amer. Math. Soc. **10**, no. 3, 565–596, DOI 10.1090/S0894-0347-97-00237-3. MR1431829
- Ikeda, Takeshi, Shinsuke Iwao, and Toshiaki Maeno. 2018. *Peterson Isomorphism in  $K$ -theory and Relativistic Toda Lattice*, preprint. arXiv: 1703.08664.
- Lam, Thomas. 2008. *Schubert polynomials for the affine Grassmannian*, J. Amer. Math. Soc. **21**, no. 1, 259–281.
- Lam, Thomas, Luc Lapointe, Jennifer Morse, and Mark Shimozono. 2010. *Affine insertion and Pieri rules for the affine Grassmannian*, Mem. Amer. Math. Soc. **208**, no. 977.
- Lam, Thomas, Anne Schilling, and Mark Shimozono. 2010.  *$K$ -theory Schubert calculus of the affine Grassmannian*, Compositio Math. **146**, 811–852.
- Lapointe, Luc, Alain Lascoux, and Jennifer Morse. 2003. *Tableau atoms and a new Macdonald positivity conjecture*, Duke Mathematical Journal **116**, no. 1, 103–146.
- Morse, Jennifer. 2011. *Combinatorics of the  $K$ -theory of affine Grassmannians*, Advances in Mathematics.
- Panyushev, Dmitri I. 2010. *Generalised Kostka-Foulkes polynomials and cohomology of line bundles on homogeneous vector bundles*, Selecta Math. (N.S.) **16**, no. 2, 315–342.